SURAJ PAWAR

AI/ML Engineer | Agentic AI & LLM Systems | RAG | Workflow Automation

Pune, Maharashtra • +91 8767707264 • srp.official.1682@gmail.com

linkedin.com/in/suraj-rajesh-pawar-a08077220a • suraj1682.github.io

PROFESSIONAL SUMMARY

AI/ML Engineer with 1.5+ years of production experience building agentic LLM systems, enterprise workflow-automation platforms, and retrieval-augmented generation (RAG) pipelines. Delivered a multi-connector Model Context Protocol (MCP) integration layer with live OAuth 2.0 flows across 8+ enterprise services, cut LLM operating cost ~46% through on-demand context retrieval, and built production RAG systems achieving 40–55% precision gains at 99.3% uptime. Strong across Python, LangGraph, FastAPI, and the modern LLM/agent tooling stack, with a track record of diagnosing and fixing production-critical bugs through live testing against real infrastructure.

TECHNICAL SKILLS

Languages: Python, SQL

Agentic AI & LLM Orchestration: LangGraph, ReAct Agents, Model Context Protocol (MCP), Multi-Provider LLM Routing (Gemini, OpenAI, Claude, Groq), Prompt Engineering, Structured Output, Token/Cost Optimization, Self-Healing & Failure-Diagnosis Agents

RAG & Vector Search: FAISS (IVF/HNSW), Qdrant, Hybrid Search (Dense + Sparse), Semantic & Heading-Based Chunking, Vector-less / PageIndex-Style Retrieval, Embeddings (Text-Embedding-3-Large), Hallucination Reduction

ML & Deep Learning: LLMs, NLP, Transformers, CNN, LSTM, Transfer Learning, LoRA/QLoRA Fine-Tuning, Model Quantization (GGUF, llama.cpp), Vision-Language Models (Qwen-VL), PyTorch, TensorFlow, Scikit-learn

Backend & Data: FastAPI, Flask, REST APIs, OAuth 2.0 (DCR, PKCE), Webhooks, PostgreSQL, MySQL, ClickHouse, Redis, Connection Pooling, ETL, Pandas, NumPy

DevOps & Cloud: Docker, GitLab CI/CD, Git, AWS, Kaggle GPU Benchmarking, Multi-Tenant / Org-Scoped Systems

AI Tools & Platforms: Claude, Claude Code, Claude Design, Gemini, Google AI Studio, ChatGPT/GPT-4, Antigravity, Cursor, GitHub Copilot

Languages (Spoken): English, Hindi, Marathi

PROFESSIONAL EXPERIENCE

AI/ML Engineer | **Yukosa** *May 2026 – Present*

MetaFlow — Activity Agent (MCP & Connectors Integration)

- Architected and shipped a **"MCP" integration system** enabling the platform to connect *any* remote **Model Context Protocol (MCP)** server on demand — not just hardcoded connectors — via a generic plugin wired into the agent's tool-orchestration layer, reusing the existing request contract so no new API surface was needed
- Built the full **OAuth 2.0** connection flow with **Dynamic Client Registration and PKCE** — automatic client registration where the vendor supports it, static-client fallback where it doesn't — and proved it end-to-end against real workspaces for **Slack, Jira/Atlassian, GitHub, Notion, and Linear**, with live tool discovery (19–27 tools per server) and confirmation-gating on destructive actions
- Designed encrypted-at-rest credential storage running on both **PostgreSQL and MySQL**, with per-user/org scoping and the full connect / disconnect / revoke lifecycle including best-effort vendor-side token revocation
- Audited and hardened **8+ enterprise connectors** (Slack, Google Sheets, MS Excel, Jira, Confluence, Outlook, Teams, HubSpot, Snowflake, Teradata), fixing real integration failures — including a long-term-memory poisoning bug and Microsoft Graph API incompatibilities — surfaced only through live testing against real infrastructure

MetaFlow — AI Assistant (Natural-Language Workflow Builder)

- Built a **LangGraph**-based, 199-activity, connector-aware **ReAct agent** that converts plain-English requests into executable drag-and-drop automation workflows, with deterministic build output via a structured JSON contract that preserves step-to-step data-flow wiring (removing a second LLM call)
- Cut LLM operating cost **~46%** by replacing an always-injected prompt catalog with on-demand, vector-less (PageIndex-style) retrieval tools over the activity catalog and workflow-pattern library

- Migrated conversation state from in-memory to a **Postgres-backed LangGraph checkpointer**, making sessions durable across restarts and multiple server instances, with session resume/list/delete and retention cleanup
- Designed **multi-tenant org-scoping** across sessions and macros and fixed a critical JWT claim-mismatch bug that had been silently denying org-scoped data to *every* logged-in user; added per-user/per-IP **rate limiting** and closed an auth-ordering vulnerability allowing unlimited secret brute-forcing
- Made the agent **connector-aware** against a 58-connector / 52-MCP-tool directory so it recommends only real, available integrations instead of hand-writing unsafe code; built a pre-flight workflow-readiness score and an optimizer/risk-analysis engine with self-maintaining catalog coverage

Datalyon (Client Engagement)

- Led research to replace the **Gemini API** with a locally self-hosted model — building an **SLM evaluation harness** that benchmarked open models (**Gemma 12B, Qwen** variants) for natural-language-to-SQL/dashboard generation with **live SQL-execution validation against ClickHouse**, catching a model whose queries parsed but failed to execute (missed by 8 prior structural-only validation rounds)
- Designed and built Datalyon's **dashboard versioning system** from the ground up — a **three-tier scheme (major / data / minor)** that snapshots dashboard structure and underlying data, with automatic no-op detection, read-time version ranking, and a data-refresh checkpoint that captures pre-refresh state; also mapped the engine's **ClickHouse-to-PostgreSQL** portability path for the planned migration
- Reduced API latency by migrating database access to **connection pooling**, eliminating per-request connection-setup overhead on high-traffic endpoints

Machine Learning Engineer | **The Strelema** *Mar 2025 – May 2026*

- Designed and deployed a production RAG system using **FAISS + Qdrant** hybrid search, improving retrieval precision **40–55%** and cutting latency from **2.5s to 300ms** via HNSW indexing, metadata filtering, and heading-based chunking
- Reduced LLM hallucinations by **25%** through a heading-based chunking strategy; integrated **Redis** for multi-turn session state, improving responsiveness by **30%**
- Automated ingestion for **5,000+ documents** with ETL pipelines and validation, maintaining **99.3% uptime** on **AWS** with continuous deployment and API monitoring
- Built a multilingual political sentiment-analysis pipeline (Marathi/Hindi/English) on **100K+** social-media comments, reaching **90.8% accuracy** with a fine-tuned L3Cube BERT model plus translation/transliteration preprocessing

Machine Learning Research Intern | **College of Engineering Pune (COEP)** *Jun 2023 – Dec 2023*

- Researched audio signal detection and source separation using deep learning — built training pipelines for audio classification with spectrogram-based feature extraction, and contributed to dataset preprocessing and model evaluation

KEY PROJECTS

RAG Search Optimization Framework | *FAISS · Qdrant · Python*

- Benchmarked retrieval across multiple embedding models, chunking strategies, and FAISS IVF/HNSW configurations for large-scale document retrieval, achieving a **35% improvement** in context relevance through vector-index tuning

Deepfake Detection Pipeline | *CNN + LSTM · OpenCV · FaceNet · Python*

- Built a hybrid CNN+LSTM architecture for temporal–spatial deepfake detection (CNN spatial features, LSTM temporal modeling), with MTCNN face alignment and multi-source data augmentation to improve generalization

EDUCATION

B.Tech, Mechanical Engineering — MIT Academy of Engineering, Pune | CGPA: 8.53/10

CERTIFICATIONS

Microsoft Azure Fundamentals (AZ-900, 900/1000) • Microsoft Azure Data Fundamentals (DP-900, 700/1000) • PCAP – Python Certified Associate Programmer • Neural Networks & Deep Learning (deeplearning.ai) • Generative AI: Applications

ACHIEVEMENTS

Winner – Machine Learning Quiz (1st of 200+ participants) • Finalist – Datathon (Equilibrium) • Bronze Badge – International Youth Mathematics Challenge